Consider a disk queue with requests for I/O to blocks on cylinders 98,183, 41, 122, 14, 124, 65, 67. The SSTF scheduling algorithm is used. The headis initially at cylinder number 53. The cylinders are numbered from 0 to 199.Write a Program to calculate the total head movement (in number of cylinders) incurred while servicing these requests.

Code:

/c/Users/john/OneDrive/Desktop/OSY_CD25D005

GNU nano 8.7

```c
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int request[] = {98,183,41,122,14,124,65,67};
    int n = 8;
    int head = 53;
    int total = 0;
    int visited[8] = {0};

    for(int i = 0; i < n; i++)
    {
        int min = 9999;
        int index = -1;

        for(int j = 0; j < n; j++)
        {
            if(visited[j] == 0)
            {
                int dist = abs(head - request[j]);
                if(dist < min)
                {
                    min = dist;
                    index = j;
                }
            }
        }

        visited[index] = 1;
        total += min;

        printf("Head moves from %d to %d = %d\n", head, request[index], min);

        head = request[index];
    }

    printf("\nTotal Head Movement = %d\n", total);

    return 0;
}
```

Output:

```
/c/Users/john/OneDrive/Desktop/OSY_CD25D005

john@DESKTOP-JAOTCLF MSYS ~
$ cd /c/Users/john/OneDrive/Desktop/OSY_CD25D005

john@DESKTOP-JAOTCLF MSYS /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ nano sstf.c

john@DESKTOP-JAOTCLF MSYS /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ gcc sstf.c -o sstf

john@DESKTOP-JAOTCLF MSYS /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ ./sstf
Head moves from 53 to 41 = 12
Head moves from 41 to 65 = 24
Head moves from 65 to 67 = 2
Head moves from 67 to 98 = 31
Head moves from 98 to 122 = 24
Head moves from 122 to 124 = 2
Head moves from 124 to 183 = 59
Head moves from 183 to 14 = 169

Total Head Movement = 323

john@DESKTOP-JAOTCLF MSYS /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$
```